Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

1)	Discuss how the number of edges in a simple graph and its complement can be related, using an example with a specific number of vertices.	**3**	(CO1, L2)
2)	Discuss how graph theory can be applied to scheduling final exams at a university, focusing on the concept of graph coloring.	**3**	(CO2, L2)
3) a)	Examine and calculate the number of internal vertices in a full 4-ary tree with 85 vertices.	**3**	(CO3, L4)
	OR		
b)	Examine and differentiate between an Eulerian graph and a Hamiltonian graph. Provide an example of each type.	**3**	(CO3, L4)
4)	Determine the number of edges in a simple graph with 8 vertices where each vertex has a degree of 3.	**3**	(CO1, L3)
5)	Define a spanning tree and construct one for the graph where = (,) with = {a, b, c, d, e} and = {{a, b}, {b, c}, {c, d}, {d, e}, {e, a}}.	**3**	(CO2, L3)
6)	Determine the number of edges in a simple graph with 8 vertices if every vertex has a degree of 3.	**3**	(CO3, L2)
7)	Determine the number of edges in a simple graph with 8 vertices if each vertex has a degree of 3. Demonstrate your answer.	**3**	(CO1, L3)
8) a)	Construct a Huffman code for the following symbols with their respective frequencies and determine the average number of bits used to encode each symbol. A: 30%, B: 25%, C: 15%, D: 10%, E: 20%. Explain your steps.	**3**	(CO2, L3)
	OR		
b)	Determine the number of edges in a simple graph with 8 vertices where each vertex has a degree of 3. Demonstrate your answer.	**3**	(CO2, L3)

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9)	Construct a spanning tree for the graph G given below and state its number of edges. Vertices: A, B, C, D, E, F Edges: AB, AC, BD, BE, CD, DE, EF	**3**	(CO3, L3)
10)	Describe a simple graph and explain how to determine the number of edges in such a graph given its number of vertices.	**3**	(CO1, L2)

Section B **7*3=21**

11)	Analyze the role of graph theory in designing efficient scheduling systems using (i) graph colorings and (ii) index structures.	**7**	(CO2, L4)
12) a)	Discuss the practical applications of graph theory in scheduling, focusing on two scenarios: (i) organizing a final exam schedule for a university and (ii) optimizing shift schedules for a hospital. Provide specific examples to illustrate your answers.	**7**	(CO3, L2)
	OR		
b)	Discuss the concepts of Eulerian and Hamiltonian paths/cycles with examples. Furthermore, explain how these concepts are relevant to solving real-world problems such as network routing in computer science.	**7**	(CO3, L2)
13)	Discuss the concept of a simple graph and illustrate your explanation by determining whether two given graphs are isomorphic. Additionally, prove that in any simple graph with an even number of vertices, there must exist at least one pair of non-adjacent vertices.	**7**	(CO1, L2)

Section C **7*4=28**

14)	Discuss the concepts of Eulerian and Hamiltonian graphs. Provide an example for each type of graph, including one that is Eulerian but not Hamiltonian. Additionally, explain the historical significance of the Konigsberg bridge problem in relation to these concepts.	**7**	(CO2, L2)
15)	Discuss the concept of a spanning tree and illustrate your explanation by finding a spanning tree for the following graphs: (i) A complete graph with 4 vertices ($_4$), and (ii) A wheel graph with 5 vertices ($_5$). Additionally, compare and contrast a spanning tree with a general tree structure.	**7**	(CO3, L2)
16) a)	Determine the number of edges in a simple graph with 8 vertices where each vertex has an odd degree. Also, construct a possible configuration for such a graph using adjacency matrix representation. (i) Explain your reasoning behind the number of edges. (ii) Provide the adjacency matrix that represents your constructed graph.	**7**	(CO1, L3)
	OR		
b)	A simple graph G has 8 vertices and is known to have an even number of edges. If one of its subgraphs H contains exactly half the number of edges as in G, determine the possible range for the number of edges in H. Also, prove that a simple graph with an odd number of vertices cannot have all its vertices of even degree.	**7**	(CO1, L3)
17)	Determine the number of internal and external vertices in a full 5-ary tree with 2048 vertices, and demonstrate that the sum of the degrees of all vertices equals twice the number of edges in this tree.	**7**	(CO2, L3)

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Section D

7*3=21

18)	Analyze the relationship between the number of edges in a simple graph and its complement. (i) Given a simple graph G with 7 vertices, determine if it is possible for G to have exactly half as many edges as its complement. Justify your answer. (ii) Construct an example of two non-isomorphic simple graphs each having 4 vertices and 5 edges, where one graph has all vertices of even degree while the other does not. Explain your choice of vertices and edges in constructing these graphs.	**7**	(CO3, L4)
19) a)	A full 4-ary tree has 65 vertices. Determine (i) how many internal vertices it has, and (ii) if this tree were to be transformed into a full binary tree with the same number of vertices, what would be the minimum height required? Justify your answers.	**7**	(CO1, L3)
	OR		
b)	Construct an Eulerian graph with at least 6 vertices that is not Hamiltonian. Then, determine whether a given subgraph of this graph is also Eulerian or Hamiltonian. Provide a clear explanation for your answers.	**7**	(CO1, L3)
20)	Construct a spanning tree for the given undirected graph G = (V, E), where V = {A, B, C, D, E} and E = {(A, B), (B, C), (C, D), (D, E), (E, A), (B, D)}. Additionally, prove that your constructed spanning tree is indeed a spanning tree by demonstrating the following: (i) it includes all vertices of G, (ii) there are no cycles in the tree, and (iii) it connects all the vertices with the minimum possible number of edges.	**7**	(CO2, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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